

Ethnopharmacological communication

Effects of Okchun-San, a herbal formulation, on blood glucose levels and body weight in a model of Type 2 diabetes

Mun Seog Chang^a, Myung Sook Oh^a, Do Rim Kim^a, Kye Jin Jung^a, Sunmin Park^b,
Soo Bong Choi^c, Byoung-Seob Ko^d, Seong Kyu Park^{a,*}^a Department of Prescriptionology, College of Oriental Medicine, Kyung Hee University, 1 Hoegi-dong, Dongdaemun-gu, Seoul 130-701, Republic of Korea^b Department of Food and Nutrition, Hoseo University, Asan-si, Chungcheongnam-do 336-795, Republic of Korea^c Department of School of Medicine, Konkuk University, Chungju-si, Chungcheongbuk-do 380-704, Republic of Korea^d Korea Institute of Oriental Medicine, 461-24 Jeonmin-Dong, Yuseong-gu, Daejeon 305-811, Republic of Korea

Received 29 March 2005; received in revised form 10 June 2005; accepted 20 August 2005

Available online 23 September 2005

Abstract

The effective control of blood glucose is the key to preventing or reversing diabetic complications. In this study, we examined the effects of Okchun-San (OCS), a Korean herbal medicine formula, on the blood glucose levels in diabetic C57BL/KsJ *db/db* mice. Herbal medicines were prepared from dried herbs by adding Coicis semen (OCS-C) or Oryzae semen (OCS-O). The experimental animals were divided into a control group and three sample groups; the sample groups received OCS-C (200 mg/kg), OCS-O (200 mg/kg), or acarbose (5 mg/kg) orally once a day for 12 days. On days 5 and 12, the fasting blood glucose levels of vehicle-treated *db/db* mice were significantly higher than day 0, while those of OCS-C, OCS-O, or acarbose-treated mice had not increased compared to the control group ($p < 0.01$). During intraperitoneal glucose tolerance test (IPGTT), OCS-C group showed improved glucose sensitivity compared to the control group. These results suggest that OCS-C can be useful as an anti-diabetic therapy for Type 2 diabetic patients by lowering the blood glucose level.

© 2005 Elsevier Ireland Ltd. All rights reserved.

Keywords: Type 2 diabetes; *db/db* mice; Okchun-San (OCS); Anti-hyperglycemia

1. Introduction

Diabetes mellitus is a major public health problem and affects ~5% of the population (Taylor, 1999). There are two common forms of diabetes: Type 1, or insulin-dependent, diabetes mellitus (IDDM) and Type 2, or non-insulin-dependent, diabetes mellitus (NIDDM) (Xie et al., 2002). Type 1 diabetes is caused by the autoimmune destruction of pancreatic β cells, leading to a near total deficiency in insulin secretion. Type 2 diabetes is the most common form of diabetes, accounting for >90% of patients. It is caused by a resistance to the action of insulin combined with a deficiency in insulin secretion (DeFronzo, 1997). The two types of diabetes have distinct causes, but two common features: hyperglycemia and various complications, which cause enormous human suffering in the form of blindness, kidney failure, amputations, and increased risk of coronary artery

disease and stroke (Taylor, 1999). Therefore, the effective control of blood glucose is the key to preventing or reversing diabetic complications and improving the quality of life for both Type 1 and Type 2 diabetic patients (DCCT Research Group, 1993; DeFronzo, 1999).

The available drugs for Type 2 diabetes have a number of limitations, such as adverse effects and high rates of secondary failure. Therefore, in recent decades, many researchers have sought new anti-diabetic remedies, which are relatively free from side effects, including medicinal herbs. As a complementary or alternative approach, medicinal herbs with anti-hyperglycemic activities are increasingly being used in the treatment of diabetes (Xie et al., 2003).

Traditional medicines have been used and case reports have indicated improvements in their quality of life as well as prolongation of life in the diabetic patients, the bio-activity of these medicines is not documented though (Bailey and Day, 1989). Generally, in traditional medicine in Asian countries, many herbal drugs are combined in the form of a multi-herbal formula to enhance their functions. The herbal constituents are selected

* Corresponding author. Tel.: +82 2 961 0330; fax: +82 2 961 0536.
E-mail address: comskp@khu.ac.kr (S.K. Park).

to emphasize the therapeutic actions or to reduce the toxicity or side effects of compounds from other herbal species in the mixture (Bansky and Barolet, 1990). The anti-diabetic effects of many traditional herbs or their fractions were reported but the effects of a multi-herbal formula, used practically, on experimental diabetes have rarely been studied.

The aqueous extract of OCS-O has long been used in traditional Korean medicinal formula to treat diabetes. OCS-C is a variant formula made using Coicis semen instead of Oryzae semen in OCS-O. Although OCS-O is the original formula that has been used, we came to know that Coicis semen contains compounds, which play a role of insulin-like action and insulin sensitizer in the previous research (Kim et al., 2002). No previous pharmacological study on OCS-C or OCS-O has been carried out. In this study, we tried to evaluate and compare the effects of these two variants on the blood glucose levels and body weight changes in the Type 2 diabetic animal model.

2. Materials and methods

2.1. Composition and preparation of the Okchun-San (OCS) extracts

2.1.1. Plant materials

Trichosanthis radix (*Trichosanthes kirilowii* Max., Cucurbitaceae), Puerariae radix (*Pueraria thunbergiana* Benth., Leguminosae), Rehmanniae radix (*Rehmannia glutinosa* Libosch., Scrophulariaceae), Schizandrae fructus (*Schizandra chinensis* Baill., Magnoliaceae), Glycyrrhizae radix (*Glycyrrhiza uralensis* Fisch., Leguminosae), Coicis semen (*Coix lachryma-jobi* L. var. *mayuen* Stapf., Poaceae), and Oryzae semen (*Oryza sativa* L., Gramineae) were purchased from Wonkwang Herbal Drug Co. Ltd. (Seoul, Korea), and authenticated by Professor S.K. Park in Department of Prescriptionology, Kyunghee University.

2.1.2. Previously isolated classes of constituents

Many chemical constituents of the herbs composing these formulas have been founded. Lectins, abortifacient proteins, and saponins from Trichosanthis radix (Wong et al., 1985), isoflavones from Puerariae radix (Wang et al., 1998), iridoid glycosides and saccharides from Rehmanniae radix (Anh et al., 2003; Bian et al., 1995), lignans and essential oils from Schizandrae fructus (Wang et al., 2003; Li et al., 2003), licorice-saponins from Glycyrrhizae radix (Kitagawa et al., 1993), coixans and fatty acids from Coicis semen (Takahashi et al., 1986; Numata et al., 1994), and heterocyclic compounds from Oryzae semen (Tao et al., 2004) have been previously reported.

2.1.3. Preparation of the OCS extracts

Aqueous extracts of OCS-C and OCS-O were prepared from six dried herbs, which are Trichosanthis radix (16.0 g), Puerariae radix (8.0 g), Rehmanniae radix (8.0 g), Schizandrae fructus (8.0 g), Glycyrrhizae radix (8.0 g), and either Coicis semen (80.0 g) for OCS-C or Oryzae semen (80.0 g) for OCS-O. These mixtures were boiled with water (50 g/l) for 2 h at 100 °C and filtered. Finally, the filtered solution containing the extract was

evaporated and lyophilized. The yields of OCS-C and OCS-O extracts were 21.9 and 20.3%, respectively.

2.2. Animals and experimental protocol

Male 6-week-old *db/db* (+/+) C57BL/KsJ mice were studied. All the animals were obtained from SLC Inc., Japan. The mice were housed in a specific-pathogen-free environment with a 12-h light:12-h dark cycle at the Center for Laboratory Animal Care and Use, Kyung Hee University. Animal care and experimental procedures conformed to the “Guide for the Care and Use of Laboratory Animals” (Department of Health, Education, and Welfare, NIH Publication #78–23, 1996). Animals had free access to standard rodent pellet food (Purina Inc., Korea) and water ad libitum, except when fasted before experiments. The *db/db* mice were used at 10 weeks of age. Body weights were monitored from days 0 to 12. Before each experiment, 50 mg of dried OCS-C and OCS-O extracts was dissolved in 1 ml distilled deionized water and vortexed for 2 min at room temperature. Each mouse was given OCS-C or OCS-O solution via gastric gavage (200 mg/kg) for 12 days. As a positive control, acarbose (Glucobay; Bayer, Korea) was administered orally (5 mg/kg) for 12 days. The same volume of vehicle was administered to control mice. No noticeable adverse effects were observed in any of the animals after the administration of the extracts, drug, or vehicle.

2.3. Fasting blood glucose test

Fasting blood glucose levels were measured after a 16 h fast on days 0 (before treatment), 5 (during treatment), and 12 (last day of treatment) using a Glucose Analyzer (Roche Diagnostic Co., USA) with blood samples drawn from the tail.

2.4. Intraperitoneal glucose tolerance test (IPGTT)

Oral and intraperitoneal glucose tolerance tests were carried out on days 0 and 12 following an overnight fast (16 h). Glucose (2 g/kg) was administered via injection into the peritoneal cavity, and blood was drawn from a tail vein at 0, 30, 60, and 120 min after glucose administration.

2.5. Statistical analysis

Data are expressed as the mean $\pm$ standard deviation. Statistical significance between the groups was determined by paired *t*-test and one-way ANOVA for repeated measures. Results with $p < 0.05$ were considered statistically significant.

3. Results

3.1. Effects of OCS-C and OCS-O on fasting blood glucose levels

As expected, all the *db/db* mice had high fasting blood glucose levels on day 0. In the control group, the fasting blood glucose level increased significantly from 270.8 ± 94.0 mg/dl

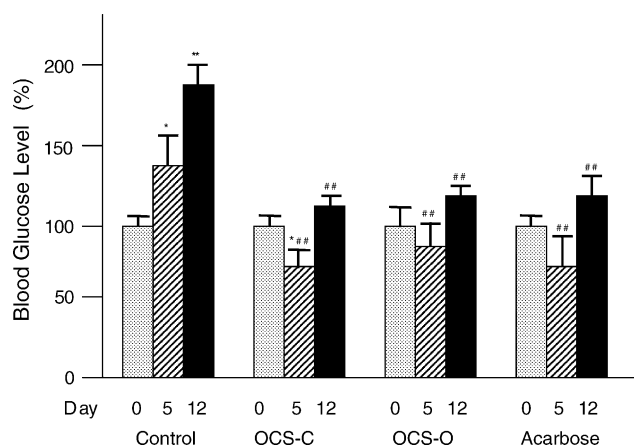


Fig. 1. Effects of the herb extracts on the blood glucose concentrations in *db/db* mice. In the vehicle ($n = 6$) and treatment ($n = 6$) groups, the mean blood glucose level on day 0 was normalized to 100%. * $p < 0.05$, ** $p < 0.01$ compared to day 0, and *** $p < 0.01$ compared to the control group.

on day 0 to 391.8 ± 47.7 mg/dl on day 5 and 498.0 ± 57.0 mg/dl on day 12 ($p < 0.05$ and < 0.01 compared to day 0, respectively). Fig. 1 shows the percentage change in the fasting blood glucose levels after treatment, with day 0 levels normalized to 100%. On days 5 and 12, the fasting blood glucose levels of the OCS-C, OCS-O, and acarbose group have not increased compared to the control group ($p < 0.01$).

3.2. Effects of OCS-C and OCS-O on the glucose tolerance test

On days 0 and 12 of the OCS-C, OCS-O, and acarbose treatments, glucose tolerance was examined using the IPGTT method. On day 0, the hyperglycemia in *db/db* mice was exacerbated by the intraperitoneal glucose load, and glucose levels failed to return to baseline after 120 min. Compared with vehicle-treated mice, mice treated with OCS-C showed a remarkable improvement in overall glucose response after 12 days (Fig. 2). In contrast, overall glucose responses in mice of

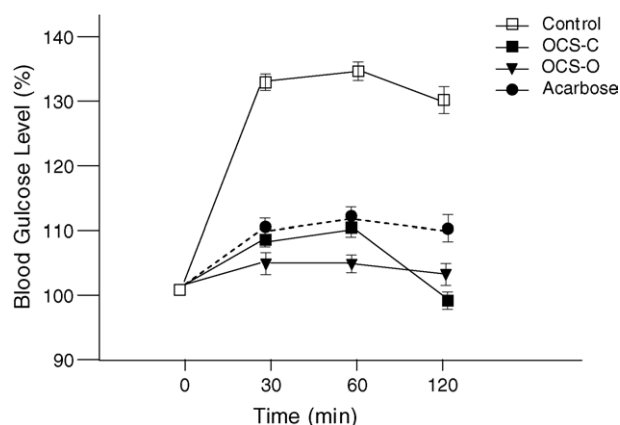


Fig. 2. Intraperitoneal glucose tolerance test in *db/db* mice after 12 days treatment with the herb extracts. All the animals were dosed with glucose (2 g/kg, i.p.). In the vehicle ($n = 6$) and treatment ($n = 6$) groups, the mean blood glucose level at 0 min was normalized to 100%. All treated groups showed a significantly higher rate of glucose disposal at 30, 60, and 120 min.

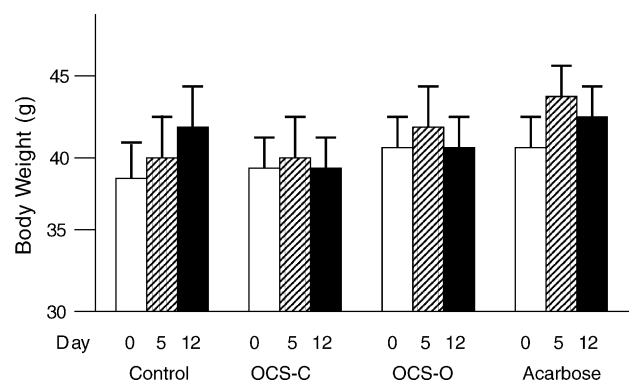


Fig. 3. Effects of the herb extracts on body weight in *db/db* mice. Body weight tended to increase from days 0 to 12 in the vehicle and acarbose (5 mg/(kg day)) treatment groups. After 12 days of treatment with OCS-C or OCS-O (200 mg/kg), body weight was measured with no exchange.

the OCS-O and acarbose treatment groups were not significantly different between days 0 and 12.

3.3. Effects of OCS-C and OCS-O on body weight change

The control group had a tendency to increase in body weight, whereas the OCS-C and OCS-O groups did not. Fig. 3 shows that the body weight of vehicle-treated *db/db* mice tended to increase from days 0 to 12 (39.4 ± 2.5 , 40.3 ± 2.4 , and 41.1 ± 2.6 g on days 0, 5, and 12, respectively). In contrast, body weight had the tendency to decrease in the OCS-C group (39.6 ± 1.9 , 39.7 ± 1.9 , and 39.3 ± 1.9 g on days 0, 5, and 12, respectively) and slightly in the OCS-O group (40.3 ± 1.6 , 40.8 ± 1.8 , and 40.1 ± 2.6 g on days 0, 5, and 12, respectively). Body weight tended to increase in the acarbose group (40.2 ± 3.3 , 42.1 ± 3.0 , and 41.5 ± 2.7 g on days 0, 5, and 12, respectively).

4. Discussion and conclusions

In this study, the anti-diabetic potential of the Korean medicinal formulations OCS-C and OCS-O were compared with that of acarbose in an animal model. Acarbose is a major oral anti-diabetic drug, which inhibits carbohydrate absorption and is in current use. We investigated the anti-hyperglycemic effect and body weight reduction activity of the herb extracts in the C57BL/KsJ strain of mice. In this model, the diabetes *db* gene mutation occurred spontaneously (Coleman, 1978). The *db/db* mice have characteristics similar to human Type 2 diabetes including obesity, hyperglycemia, and extreme insulin resistance. The mice are obese and hyperinsulinemic up to 1 month of age, then insulin resistance worsens with the appearance of hyperglycemia. The insulin resistance in mature *db/db* mice is a manifestation of hyperglycemia and remaining hyperinsulinemia that occurs between 2 and 3 months of age (Kodama et al., 1994). We demonstrated that both OCS-C and OCS-O showed hypoglycemic activity in this model. We started testing *db/db* mice at 10 weeks of age. During experimental period, fasting blood glucose levels in the control group have increased significantly. But those of experimental

groups did not increase after 5 and 10 days treatment with 200 mg/kg of OCS-C or OCS-O extract, exhibited an efficacy similar to that of acarbose (Fig. 1). And OCS-C reduced the fasting blood glucose level in the glucose tolerance test (Fig. 2).

OCS-C and OCS-O are composed of several herbs having hypoglycemic activities. The non-dialyzable portion of the water extract of *Trichosanthis radix* was found to reduce the plasma glucose level in mice. The main glycan of *Trichosanthis radix*, trichosan A, also exhibited activity in alloxan-induced hyperglycemic mice (Hikino et al., 1989). Oligosaccharide of *Rehmanniae radix* exerted a significant hypoglycemic effect in normal and alloxan-induced diabetic rats (Zhang et al., 2004). Glycyrrin, one of the main PPAR- γ ligands of *Glycyrrhizae radix*, significantly decreased the blood glucose levels of genetically diabetic KK-A(y) mice (Kuroda et al., 2003). Coicis semen was found to have hypoglycemic activity of coixans and glycans (Takahashi et al., 1986). These constituents of the formulas may be responsible for its hypoglycemic effect.

Interestingly, no statistically significant weight change was observed in *db/db* mice on days 5 and 12, while body weight tended to increase in the control group (Fig. 3). Perhaps, the anti-hyperglycemic effect of OCS-C extract we observed is independent of the body weight reduction effect. Likewise, acarbose treatment provided only a very modest and insignificant benefit for weight reduction, whereas it decreased insulin requirements as an α -glucosidase inhibitor (Hauner et al., 2001). The therapeutic strategy for the obese diabetic patient includes not only promoting weight loss through a low-calorie diet and exercise but also improving blood glucose control through agents that decrease insulin resistance or insulin needs (Scheen, 2003).

Furthermore, we measured glycosylated hemoglobin (HbA1c), which represents the number of glucose molecules attached to hemoglobin, in red blood cells (data not shown). HbA1c is also widely used in screening for diabetes mellitus and glucose intolerance (Shafir, 1992; Xie et al., 2002). We speculate that the lack of improvement in the HbA1c levels in any of the treated animals was partly owing to the short treatment period and low dose. HbA1c is reported to be an objective assessment of the integrated blood glucose concentration over the preceding 2–3 months (DeFronzo, 1999; Chakrabarti et al., 2002; Xie et al., 2003).

Recently, ethnic research and natural product screening have led to advances in anti-diabetic therapeutics in a Type 2 diabetes model. The effect of *Helicteres isora* on insulin-sensitizing and hypolipidemic activity has been examined (Chakrabarti et al., 2002). In addition, *Gymnema yunnanense* extract and American ginseng berry extract were shown to exhibit anti-hyperglycemic activity in a diabetes model (Xie et al., 2002, 2003). Similarly, we demonstrated that the administration of OCS-C or OCS-O reduced high blood glucose levels. This is the first report about OCS-C, whereas some anti-diabetic effects of OCS-O have been studied (Koh and Kim, 2002; Oh and Choi, 1990). But the diabetic models they used were Type 1 acutely induced by chemicals like alloxan or streptozotocin. In the Type 2 diabetes

model we carried out, OCS-C showed more improved hypoglycemic effects on diabetes statistically.

In conclusion, OCS-C shows the hypoglycemic effects in the *db/db* mouse model. Further detailed researches of mechanisms and clinical trials are in progress.

Acknowledgment

This study was supported by a grant of the Oriental Medicine R&D Project, the Ministry of Health & Welfare, Republic of Korea (02-PJ9-PG1-CO02-0002).

References

- Anh, N.T., Sung, T.V., Franke, K., Wessjohann, L.A., 2003. Phytochemical studies of *Rehmannia glutinosa* rhizomes. *Pharmazie* 58, 593–595.
- Bailey, C.J., Day, C., 1989. Traditional plant medicines as treatments for diabetes. *Diabetes Care* 12, 553–564.
- Bansky, D., Barolet, R., 1990. Chinese Herbal Medicine Formulas and Strategies. Eastland Press, Seattle, pp. 7–8.
- Bian, B., Wang, H., Ni, M., 1995. Determination of total saccharide and several main saccharides in *Rehmannia glutinosa* Libosh, and its processed products. *Zhongguo Zhong Yao Za Zhi* 20, 469–471, 511–512.
- Chakrabarti, R., Vikramadithyan, R.K., Mullangi, R., Sharma, V.M., Jagad-heshan, H., Rao, Y.N., Sairam, P., Rajagopalan, R., 2002. Antidiabetic and hypolipidemic activity of *Helicteres isora* in animal models. *Journal of Ethnopharmacology* 81, 343–349.
- Coleman, D.L., 1978. Obese and diabetes: two mutant genes causing diabetes–obesity syndromes in mice. *Diabetologia* 14, 141–148.
- DeFronzo, R.A., 1997. Pathogenesis of type 2 diabetes: metabolic and molecular implications for identifying diabetes genes. *Diabetes Reviews* 5, 177–269.
- DeFronzo, R.A., 1999. Pharmacologic therapy for type 2 diabetes mellitus. *Annals of Internal Medicine* 131, 281–303.
- Diabetes Control Complications Trial Research Group, 1993. The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. *New England Journal of Medicine* 329, 977–986.
- Hauner, H., Petzinna, D., Sommerauer, B., Toplak, H., 2001. Effect of acarbose on weight maintenance after dietary weight loss in obese subjects. *Diabetes, Obesity and Metabolism* 3, 423–427.
- Hikino, H., Yoshizawa, M., Suzuki, Y., Oshima, Y., Konno, C., 1989. Isolation and hypoglycemic activity of trichosans A, B, C, D, and E: glycans of *Trichosanthes kirilowii* roots. *Planta Medica* 55, 349–350.
- Kim, J., Choi, Y., Ju, Y., Park, S., Lee, M., Kim, H.K., Kim, H.J., Ko, B., 2002. Effect of insulin-like action and insulin sensitizing on 3T3-L1 adipocytes from Coicis semen. *Journal of Korean Oriental Medicine* 23, 83–91.
- Kitagawa, I., Hori, K., Sakagami, M., Zhou, J.L., Yoshikawa, M., 1993. Saponin and saponenol. XLVIII. On the constituents of the roots of *Glycyrrhiza uralensis* Fischer from northeastern China. (2). Licorice-saponins D3, E2, F3, G2, H2, J2, and K2. *Chemical & Pharmaceutical Bulletin* 41, 1337–1345.
- Kodama, H., Fujita, M., Yamaguchi, I., 1994. Development of hyperglycaemia and insulin resistance in conscious genetically diabetic (C57BL/KsJ-db/db) mice. *Diabetologia* 37, 739–744.
- Koh, J.B., Kim, J.Y., 2002. Effect of okcheonsan on blood glucose, lipid and protein bevels in streptozotocin-induced diabetic female rats. *Journal of the Korean Society of Food Science and Nutrition* 31, 284–289.
- Kuroda, M., Mimaki, Y., Sashida, Y., Mae, T., Kishida, H., Nishiyama, T., Tsukagawa, M., Konishi, E., Takahashi, K., Kawada, T., Nakagawa, K., Kitahara, M., 2003. Phenolics with PPAR- γ ligand-binding activity obtained from licorice (*Glycyrrhiza uralensis* roots) and ameliorative effects of glycyrrin on genetically diabetic KK-A(y) mice. *Bioorganic and Medicinal Chemistry Letters* 13, 4267–4272.

- Li, X.N., Cui, H., Song, Y.Q., Liang, Y.Z., Chau, F.T., 2003. Analysis of volatile fractions of *Schisandra chinensis* (Turcz.) Baill. using GC–MS and chemometric resolution. *Phytochemical Analysis* 14, 23–33.
- Numata, M., Yamamoto, A., Moribayashi, A., Yamada, H., 1994. Antitumor components isolated from the Chinese herbal medicine *Coix lachryma-jobi*. *Planta Medica* 60, 356–359.
- Oh, J.S., Choi, D.Y., 1990. Effects of oak Cheun-san on metabolites of alloxan-intoxicated rats. *Korean Journal of Oriental Pathology* 5, 77–88.
- Scheen, A.J., 2003. Current management strategies for coexisting diabetes mellitus and obesity. *Drugs* 63, 1165–1184.
- Shafir, E., 1992. Animal models of non-insulin-dependent diabetes. *Diabetes Metabolism* 8, 180–208.
- Takahashi, M., Konno, C., Hikino, H., 1986. Isolation and hypoglycemic activity of coixans A, B and C, glycans of *Coix lachryma-jobi* var. mayuen seeds. *Planta Medica* 53, 64–65.
- Tao, S., Wang, L., Zhang, J., Yuan, S., Mao, Z., 2004. Analysis of volatile heterocyclic compounds in glutinous rice after high temperature fluidization by GC/MS. *Se Pu* 22, 189.
- Taylor, S.I., 1999. Deconstructing type 2 diabetes. *Cell* 97, 9–12.
- Wang, C.Y., Huang, H.Y., Kuo, K.L., Hsieh, Y.Z., 1998. Analysis of *Puerariae radix* and its medicinal preparations by capillary electrophoresis. *Journal of Chromatography A* 802, 225–231.
- Wang, Y.H., Gao, J.P., Chen, D.F., 2003. Determination of lignans of *Schisandra* medicinal plants by HPLC. *Zhongguo Zhong Yao Za Zhi* 28, 1155–1160.
- Wong, C.M., Yeung, H.W., Ng, T.B., 1985. Screening of *Trichosanthes kirilowii*, *Momordica charantia* and *Cucurbita maxima* (family Cucurbitaceae) for compounds with antilipolytic activity. *Journal of Ethnopharmacology* 13, 313–321.
- Xie, J.T., Aung, H.H., Wu, J.A., Attele, A.S., Yuan, C.S., 2002. Effects of American ginseng berry extract on blood glucose levels in ob/ob mice. *American Journal of Chinese Medicine* 30, 187–194.
- Xie, J.T., Wang, A., Mehendale, S., Wu, J., Aung, H.H., Dey, L., Qiu, S., Yuan, C.S., 2003. Anti-diabetic effects of *Gymnema yunnanense* extract. *Pharmacological Research* 47, 323–329.
- Zhang, R., Zhou, J., Jia, Z., Zhang, Y., Gu, G., 2004. Hypoglycemic effect of *Rehmannia glutinosa* oligosaccharide in hyperglycemic and alloxan-induced diabetic rats and its mechanism. *Journal of Ethnopharmacology* 90, 39–43.